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STUDENT A & STUDENT B
MASTER'S THESIS
DEPARTMENT OF ELECTRICAL AND INFORMATION TECHNOLOGY
FACULTY OF ENGINEERING | LTH | LUND UNIVERSITY

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An unofficial Typst template for degree project at Electrical and Information Technology at LTH, made by Lucas Ekholm (E22)

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Frontmatter

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AC Actually Confusing

CODE Computers Only Do Exactly (what you tell them)

EIT Engineering Is (never) Trivial

EMC Engineers' Major Complaint

IC Infinite Confusion

LED Let's Emit Dazzle

MOSFET Mostly Our Small Friend Enabling Transistor

PING Packet Is Now Going

> "To mature as an electrical engineer is to realize everything is about impedances."
>
> — Markus Törmänen[1], spring 2025

[1] Translated. Original quote in Swedish.

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Chapter 1

Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

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$$\operatorname{div} \boldsymbol{A} \coloneqq \lim_{\Delta v \to 0} \frac{\oiint_S \boldsymbol{A} \cdot d\boldsymbol{s}}{\Delta v} \tag{1.1}$$

$$\nabla \times \boldsymbol{B} = \mu_0 \left(\boldsymbol{J} + \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \right) \tag{1.2}$$

The equations above are (1.1) and (1.2). This is Chapter 1 but below is Section 1.1 and below that is Section 1.1.1. This chapter can be found on page 1.

1.1 Test section

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1.1.1 A section which won't appear in the outline on page iv.

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1.1.1.1 These sections keep going!

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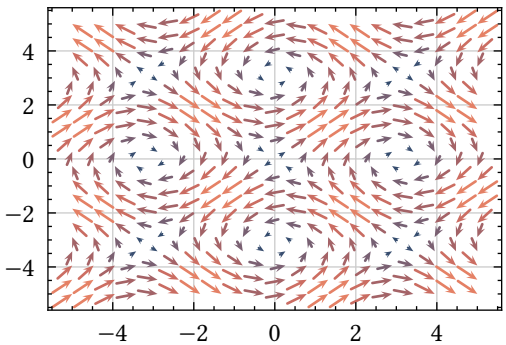

Fig. 1.1: An example figure using the package Lilaq. [1]

1.2 Another test section

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.[2]

[2] https://www.lth.se

Fig. 1.2: A figure with a longer caption. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

Chapter 2

New Chapter with a Long Title that Spanns over More Than One Line

Fig. 2.1: A figure together with another figure.

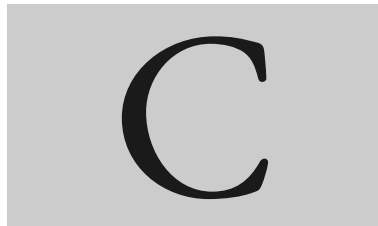

Fig. 2.2: This figure can have a long caption and still fit if done this way. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

Above Fig. 2.1 is next to Fig. 2.2. By using the `#flexCaption(long caption, short caption)` function you can make flexible captions. You can also make longer/shorter headings! See Chapter 2 on page iv.

2.1 New section

$$F_n = F_{n-1} + F_{n-2}, \qquad \begin{cases} F_0 = 0 \\ F_1 = 1 \end{cases} \tag{2.1}$$

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Table 2.1: The Fibonacci sequence up to n = 15.

F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}
1	1	2	3	5	8	13	21	34	55	89	144	233	377	610

Chapter 3

A word on Typst referencing

On page 9 you can find links. [2] provides a guide for past LaTeX users on the differences between the two typesetting languages, including what's native or not native, which macros map to which function etc. [3] and [4] links to information on how to cite and reference different sources. Typst includes support for both its native YAML-based format and BibTeX. [5] gives a link to the Typst Universe. On this page you can find packages which aid in your workflow, both in terms of styling and automation. For example Fig. 1.1 was made using lilaq, a package used to make graphs inside the Typst environment. Another good package is Zap which is used to make circuit diagrams much like TikZ. [6]

For documentation on all functions, markup- and styling commands, see [7].

References

[1] "Lilaq - Typst Universe." [Online]. Available: https://typst.app/universe/package/lilaq/

[2] "Guide for LaTeX users." [Online]. Available: https://typst.app/docs/guides/guide-for-latex-users/

[3] "Cite function." [Online]. Available: https://typst.app/docs/reference/model/cite/

[4] "Bibliography function." [Online]. Available: https://typst.app/docs/reference/model/bibliography/

[5] "Typst Universe." [Online]. Available: https://typst.app/universe/

[6] "Zap - Typst Universe." [Online]. Available: https://typst.app/universe/package/zap

[7] "Typst Documentation." [Online]. Available: https://typst.app/docs

Appendix A

Backmatter

```c
/* Function to sort array using insertion sort */
void insertionSort(int arr[], int n)
{
    for (int i = 1; i < n; ++i) {
        int key = arr[i];
        int j = i - 1;

        /* Move elements of arr[0..i-1], that are
           greater than key, to one position ahead
           of their current position */
        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j = j - 1;
        }
        arr[j + 1] = key;
    }
}
```

Fig. A.1: Code snippet for insertionsort written in C.

Fig. A.1: This is the last figure.

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